DO NOW — RECALL

# Database tables, entities and keys

Name: _____ Class: _____ Date: _____

**1. Define INNER JOIN.**

**2. What is the purpose of a primary key?**

**3. Which SQL clause filters rows?**

**4. Insert 12 into BST with root 10 and right child 15. State its final position.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Combines matching table rows.

**2. Model answer**

Uniquely identifies each row.

**3. Model answer**

WHERE.

**4. Model answer**

12>10→right to15; 12<15→left child of15.

**Vocabulary — know these terms**

entity — Type of identifiable object.

attribute — Descriptive field of an entity.

primary key — Unique row identifier.

foreign key — Reference to key in another table.

index — Search aid on selected columns.





# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

**Guided 1 — original CHECK**

Model answer

Explain the reasoning and justify the output.

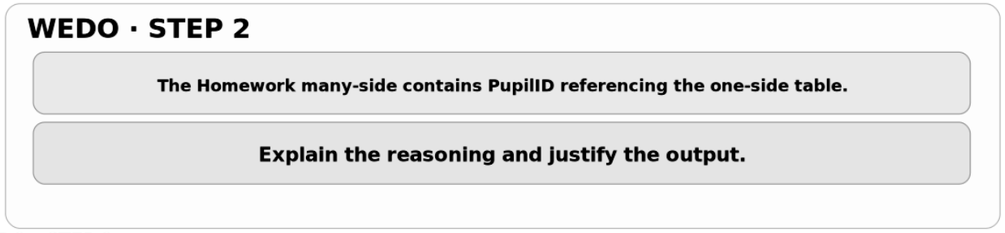
IMPROVE: Explain one correction to your own reasoning.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A pupil may submit several Homework rows. Choose PK and FK.**

The Homework many-side contains PupilID referencing the one-side table.




WEDO · STEP 2





NEW LEARNING

# Read, annotate and explain

### Understand

An entity corresponds to a distinguishable object type; a relational table stores rows of its attributes.

### Apply

A primary key uniquely identifies each row; foreign keys express references from related tables.

### Evaluate

A secondary/index key can accelerate searching without necessarily being unique. One-to-many relationships use a foreign key on the many side.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

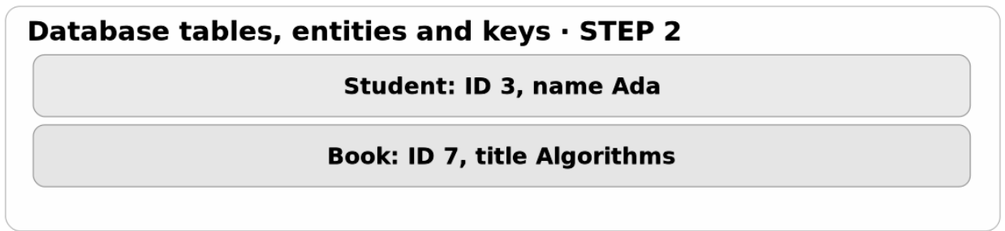
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



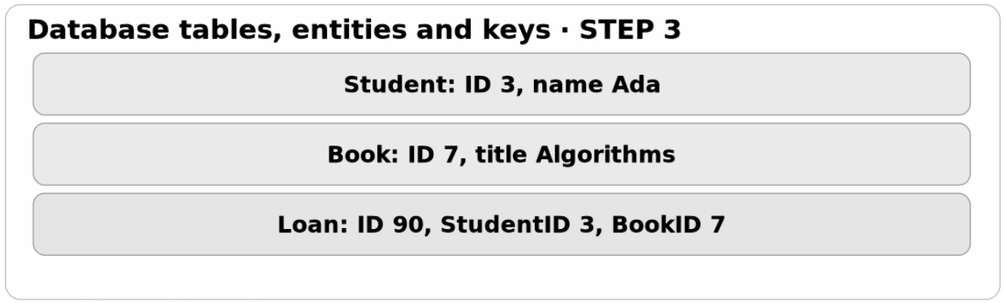

Database tables, entities and keys · STEP 1

**New Learning**




Database tables, entities and keys · STEP 2

**New Learning**




Database tables, entities and keys · STEP 3





**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A foreign key must always be unique across every row in its table.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Explain why storing every loan and every pupil detail in a single repeated row causes anomalies. [6]

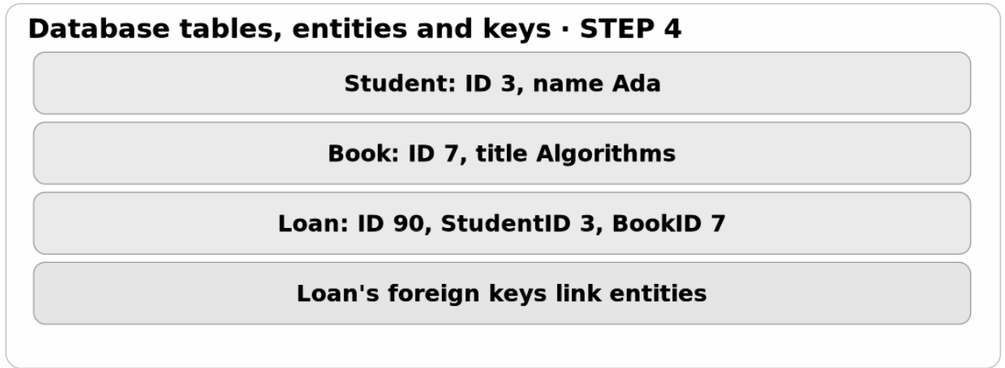




# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Database tables, entities and keys · STEP 4

I DO — WORKED EXAMPLE

# Follow the worked method

**Identify primary and foreign keys in Customer(CustomerID, Name) and Order(OrderID, CustomerID).**

### Predict first

Orders have their own identity and store a reference to the customer who placed them.

### Model answer

Explain the reasoning and justify the output.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Define entity, attribute, primary key, foreign key. [4]

### Question 2

For Student, Book, Loan given above, state PKs and both FKs. [6]

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Predict first The Homework many-side contains PupilID referencing the one-side table.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.





# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Identify primary and foreign keys in Customer(CustomerID, Name) and Order(OrderID, CustomerID).**

Orders have their own identity and store a reference to the customer who placed them.




IDO · STEP 1

**Identify primary and foreign keys in Customer(CustomerID, Name) and Order(OrderID, CustomerID).**

Orders have their own identity and store a reference to the customer who placed them.




IDO · STEP 2

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve



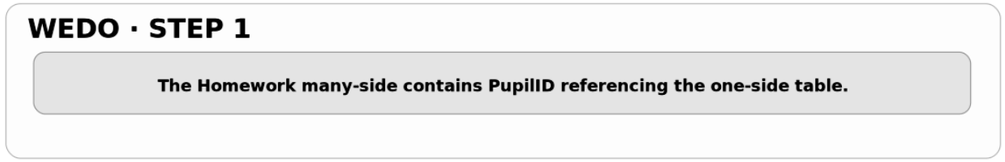


## We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A pupil may submit several Homework rows. Choose PK and FK.**

The Homework many-side contains PupilID referencing the one-side table.




WEDO · STEP 1